Getting Started and Getting Help

1. **Getting Started**
   - SageMath and CoCalc, Jupyter notebooks
2. **Getting Help**
   - personal and online
3. **Precision and Accuracy**
   - approximating numbers

MCS 320 Lecture 3
Introduction to Symbolic Computation
Jan Verschelde, 28 August 2026

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Getting Started

The essential points about this course are listed below.

1. The course web site is at
`http://homepages.math.uic.edu/~jan/mcs320`.

2. Install (and/or signup) the free open source system SageMath.
   - `https://www.sagemath.org` for source, docs, installation, etc.
   - `https://cocalc.com` for online execution.

3. The Jupyter notebooks that you will make in this course will be your most important resources.

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Getting Help

1. Office hours are Monday, Wednesday, Friday, online, or, on campus in SEO 1210, from 2pm to 3pm. We can also make an appointment.
2. SageMath is well documented and offers online help.
3. Search engines may bring you to online forums and give hints on the "How To Do ..." aspects of the course.
4. Be cautious about the output generated by the AI in cocalc ...

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Precision

Definition (precision)

The *precision* of a number system is the smallest positive number we can add to one and still make a difference.

Examples:

- A decimal number system with 3 decimal places:

$$1.00 + 0.001 = 1.00 \quad \text{and} \quad 1.00 + 0.005 = 1.01$$

  In this number system, the precision is thus $0.005 = 5 \times 10^{-3}$.

- A binary number system using 3 bits:

$$1.00 + 0.0001 = 1.00 \quad \text{and} \quad 1.00 + 0.001 = 1.01$$

  In this number system, the precision is thus $0.001 = 1 \times 2^{-3}$.

Accuracy

Definition (accuracy)

The *accuracy* of an approximation is the relative error, computed as the absolute value of the difference between the exact, nonzero value and the approximation, divided by the exact value.

Example: What is the accuracy of $22/7$ as an approximation for π?

$$\left| \frac{22/7 - \pi}{\pi} \right|$$

- Accuracy is a property of an approximation, precision concerns the number system.
- Errors should be computed in a precision at least as high as the approximation.